

Concurrency and Distributed Systems

Week 6 – Part 4: Blocking Hazards and Advanced Bridge

Dr. Mohamad Aoude

Faculty of Engineering

May 13, 2026

Part 4 Roadmap

- 1 Blocking Hazards
- 2 Advanced Patterns
- 3 Lab Wrap-Up
- 4 Week 7 Bridge

Blocking in ForkJoinPool

Pitfall

Blocking I/O inside the common `ForkJoinPool` can starve CPU-bound work.

Fork/Join assumes workers mostly compute. If workers wait on network, disk, sleep, or locks, other subtasks may not get CPU time.

Why Starvation Happens

- The pool has limited worker parallelism.
- A blocked worker cannot execute or steal work.
- Parallel streams and other common-pool tasks share those workers.
- Throughput falls even though the machine may have queued work ready.

Better Boundary

Good Fork/Join fit	Poor Fork/Join fit
CPU-bound computation Independent ranges Combine returned values Measured threshold	Blocking network calls Tasks waiting on external services Shared mutable counters One task per tiny item

Optional: CompletionService

Where it fits

`ExecutorCompletionService` returns completed tasks as they finish.

This is useful for work-as-ready workflows. It is not Fork/Join decomposition; it is a bridge toward async and fan-out/fan-in design.

Optional: FutureTask and Memoization

- FutureTask wraps a computation that can run once.
- Other threads can wait on `Future.get()`.
- A map of keys to FutureTask can avoid duplicate work.

Teaching use

Keep this as optional reading unless students have finished the main Week 6 exercises.

Week 6 Lab Wrap-Up

- Ex1: implement a correct Fork/Join sum.
- Ex2: measure and explain the break-even point.
- Ex3: diagnose parallel-stream slowdown.
- Scorecard: deterministic, independent, thresholded, no blocking I/O, fair measurement.

Week 7 Bridge

Week 7 changes the question again.

- Week 5: How do we bound task execution?
- Week 6: When should CPU work be split?
- Week 7: How do we compose waiting, timeouts, fallbacks, and async results?

Exit Check

- ① Why is blocking inside the common pool risky?
- ② Why is `CompletionService` not the same thing as `Fork/Join`?
- ③ What evidence should a Week 6 performance claim include?